

Clinical decisions we need from you

Nine questions. Most people finish in about twenty minutes.

We have built a computer system that checks whether a kidney donor and a recipient are compatible, and that looks for possible paired exchanges when a pair is not directly compatible. The software is finished and tested. What it does not have is your judgement.

At nine points the system currently applies a number or a rule that no clinician has agreed to. We chose provisional values so the software could be built and tested, and we have deliberately left them unchanged since — because changing them twice would invalidate the testing. Each question below is one of those points.

You do not need to know anything about the software to answer. Every question tells you what the system does today and why we are unsure about it. If a question does not have a right answer, say so — “this should be left to the clinician case by case” is a perfectly good response, and we can build that.

HOW TO USE THIS FORM

- Tick one option per question, or write your own under “Other”.
- If you disagree with each other, tick both and note who preferred what. Disagreement is useful — it tells us the decision needs a wider discussion, not a default.
- If you would rather not answer a question without more information, tick the last option and tell us what you would need.
- Nothing here is final. These values are recorded with a version number, so they can be revised later and past reports stay interpretable.

Completed by

Name

Role / speciality

Signature and date

Name

Role / speciality

Signature and date

SECTION A

The rules that stop a transplant

These three decide whether the system tells a doctor “not compatible”. They matter most.

QUESTION 1

Should the number of HLA mismatches stop a transplant?

WHAT THE SYSTEM DOES NOW

It counts mismatches at three loci — A, B and DRB1 — giving a score from 0 to 6.

At 6 out of 6 it refuses the pairing outright and reports “not compatible”.

Below 6 the count feeds into the final risk level but does not block anything.

Why we are asking. We are not confident this is right. In living donation a fully mismatched unrelated donor — a spouse, for example — is common and transplanted successfully every day, with the real immunological risk decided by antibody testing and the crossmatch rather than the raw mismatch count. As written, the system would refuse a spousal donation that your unit would accept. We have left the threshold in place rather than remove it on our own judgement.

Please tick one:

- ☐ Remove it. Mismatch count should never block a pairing on its own — report it as risk information and let antibody testing and the crossmatch decide.
- ☐ Keep a block, but not at 6. Refuse only at a mismatch count of: _____
- ☐ Keep it exactly as it is — refuse at 6 out of 6.
- ☐ Something else, or I need more information (please write below).

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QUESTION 2

Antibody strength: at what level should we stop?

WHAT THE SYSTEM DOES NOW

Antibodies against the donor's HLA are graded by MFI:

below 1,000 — not treated as a donor-specific antibody at all

1,000 – 1,999 — “weak”: flagged for review, transplant proceeds

2,000 – 4,999 — “moderate”: flagged for review, transplant proceeds

5,000 and above — **“strong”: the system refuses the pairing**

Separately, an antibody above MFI 2,000 counts towards the recipient's cPRA (how hard they are to match against the population).

Why we are asking. These cut-offs came from general reading, not from your laboratory. MFI values are not comparable between laboratories or between kits, so the right numbers depend on the assay your unit actually uses. We would rather use your thresholds than a textbook's.

Please tick one:

- ☐ The thresholds above are right — keep them.
- ☐ Use these thresholds instead (please write the MFI values below).
- ☐ Change which band stops the transplant — it should stop at: _____
- ☐ Our laboratory reports something other than MFI, or reports it differently (please describe).

Your thresholds / notes

QUESTION 3

Crossmatch: which test, and which result stops a transplant?

WHAT THE SYSTEM DOES NOW

It asks one yes/no question: “was the crossmatch positive?” Any positive result stops the transplant.

There are free-text boxes for a T-cell and a B-cell result, but the system does not read them — they are stored and displayed only.

It does not record whether the crossmatch was CDC or flow cytometry.

Why we are asking. We think this is too blunt, but it is not our call to change it. A positive T-cell CDC crossmatch and an isolated positive B-cell flow crossmatch do not mean the same thing, and many units proceed with the second. As written, the system treats them identically. We can capture whichever fields you actually use — we need to know which combination should stop a transplant and which should only raise a flag.

Please tick all that should STOP a transplant outright:

- ☐ Positive T-cell CDC crossmatch
- ☐ Positive B-cell CDC crossmatch
- ☐ Positive T-cell flow crossmatch
- ☐ Positive B-cell flow crossmatch
- ☐ Any positive result, regardless of test or cell type (i.e. leave it as it is)

And which should only be flagged for review, not blocked?

Which crossmatch method does your unit use routinely?

- ☐ CDC only
- ☐ Flow cytometry only
- ☐ Both
- ☐ Other / it varies — _____

Scoring and wording

These change what a report says, not whether a transplant is allowed.

QUESTION 4

How should we record a recipient's sensitising history?

WHAT THE SYSTEM DOES NOW

Three tick-boxes with point values: previous transplant (2.0), prior pregnancy (1.0), blood transfusion (0.5).

The points are added up and shown on the report as a "sensitisation score".

The score is displayed only. It does not change any threshold and does not affect the result.

Why we are asking. The point values were assigned by us, not derived from anything. We are also unsure the fields are right: a yes/no for pregnancy loses the number of pregnancies, and a yes/no for transfusion loses how many and how recently. If the score is not clinically meaningful, we would rather remove it than display a number that looks authoritative and is not.

Please tick one:

- ☐ Remove the score. Just list the history as facts, with no points.
- ☐ Keep the score but change the values (please write them below).
- ☐ Keep it as it is.
- ☐ Keep it, and also capture:
 - ☐ number of pregnancies ☐ number of transfusions ☐ dates of each event

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QUESTION 5

What should the risk levels be called?

WHAT THE SYSTEM DOES NOW

Every completed report ends with one of four risk levels:

Low Risk • Low-Average Risk • High-Average Risk • High Risk

A separate, older score elsewhere in the system uses: Low, Moderate, High-Moderate, High Immunological Risk.

Why we are asking. “Low-Average” and “High-Average” are not words we have seen used clinically, and having two different four-level scales in one system is confusing. We would like one set of names that reads correctly in a clinical note.

Please tick one:

- ☐ Low / Intermediate / High / Very High
- ☐ Low / Moderate / High / Very High
- ☐ Keep the current names.
- ☐ Use these names instead:

Is there any other wording in the reports that reads wrongly to you? We have already corrected a set of terms, but you will see things we cannot.

Wording to change

QUESTION 6

Highly sensitised recipients: how much should cPRA count?

WHAT THE SYSTEM DOES NOW

cPRA is grouped into three bands: under 30%, 30–60%, and above 60%.

The final risk level is worked out by adding points from the mismatch count and points from the cPRA band. Under 30% scores 0 points; 30–60% scores 1 point.

Above 60% has no agreed point value, so those reports come out with no risk level at all — they just say “proceed with caution”.

Why we are asking. This is a gap we left deliberately rather than invent a number. It affects exactly the recipients who are hardest to match, so it is the group where a missing risk level is least acceptable.

Please tick one:

- ☐ Above 60% should score 2 points (so it can reach the highest risk level).
- ☐ Above 60% should score 1 point, the same as 30–60%.
- ☐ Above 60% should score: _____
- ☐ The bands themselves are wrong. Use these instead:

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- ☐ A highly sensitised recipient should not get an automatic risk level at all — it needs a clinician's judgement each time.

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Paired exchange

When a pair is not directly compatible, the system looks for a swap with another pair.

QUESTION 7

How much HLA typing before a pair can be offered a swap?

WHAT THE SYSTEM DOES NOW

A pair can enter the exchange pool once both people have typing at A, B and DRB1.

DQ, DP and C typing are not required. If a recipient has an antibody against a locus the donor was never typed for, the system currently cannot detect it and treats the pairing as clear.

We are fixing the system so it flags this rather than staying silent. What we need from you is what the flag should do.

Why we are asking. This is a real safety question and we would rather ask than assume. An unrecognised anti-DQ antibody is a known cause of an unexpected positive crossmatch. Requiring full typing is safer but will keep some pairs out of the pool; requiring less is more inclusive but relies on the crossmatch to catch it later.

Please tick one:

- ☐ Require A, B, DRB1 and DQ typing before a pair enters the exchange pool.
- ☐ Require full typing (A, B, C, DRB1, DQ, DP).
- ☐ Keep A, B, DRB1 as the minimum, but if the recipient has an antibody against an untyped locus, warn prominently and let the coordinator decide.
- ☐ Keep A, B, DRB1 as the minimum, and block that specific swap until the missing typing is entered.
- ☐ Something else:

QUESTION 8

How long should an exchange offer stay open?

WHAT THE SYSTEM DOES NOW

A proposed swap expires automatically after 7 days if every pair has not accepted.

Once it expires the pairs return to the pool and can be matched into a different swap.

Why we are asking. We chose 7 days with no idea how long the real process takes. If approvals in practice take two to four weeks, every proposal will expire before anyone can act on it, and the system will look broken.

Please tick one:

- ☐ 7 days is right.
- ☐ It should be: _____
- ☐ It should not expire automatically at all — a coordinator should cancel it manually.
- ☐ It depends on the stage — please describe:

Roughly how long does it take, in practice, from identifying a possible swap to both units agreeing?

QUESTION 9

Are you comfortable with how we calculate cPRA?

WHAT THE SYSTEM DOES NOW

cPRA is estimated from the recipient's unacceptable antigens combined with HLA antigen frequencies measured in 714 Colombo blood donors (Grifoni et al., 2018).

It combines each antigen's frequency as if they were independent. The standard international method uses haplotype frequencies, which we do not have for a Sri Lankan population.

Our figure will therefore differ from a cPRA calculated by UNOS or Eurotransplant tools, most likely by overestimating.

Why we are asking. We think using local frequency data is the right choice — a US or European table would be worse for your patients. But it is an approximation, and the report currently presents the number without saying so. We do not want a clinician comparing it to a cPRA from elsewhere without knowing the difference.

Please tick one:

- ☐ The method is acceptable. Label it clearly as an estimate on the report.
- ☐ The method is acceptable. No extra labelling needed.
- ☐ Show it as a band only (e.g. "under 30%"), not a precise percentage.
- ☐ This needs review by an immunologist before the pilot.
- ☐ Something else:

Anything we have not asked about

These nine are the decisions we know we are missing. There will be others we have not thought to ask. If anything about how this system works, what it shows, or where it might mislead someone concerns you, it is more useful here than anywhere else.

One last thing

BEFORE THIS IS USED WITH REAL PATIENTS

We would like two or three of you to run about ten real, anonymised donor-recipient pairs through the finished system and tell us whether you agree with every verdict it produces. Nothing else gives us the same confidence.

Would you be willing to do that?

- ☐ Yes ☐ Yes, but not me personally — speak to: _____
☐ No

Thank you. Please return this form, or hand it back at the meeting. Nothing in the system will be changed on the basis of a value we chose ourselves once we have your answers.